

CONCEPT FORMATION



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6.1 INTRODUCTION

As you read in the last chapter, children learn and develop any concept through play. They acquire new knowledge, identify objects, and understand their relationships through play in an enjoyable manner, which makes learning easy and interesting. Concept formation refers to organizing experiences into general categories and classifying objects which is influenced by play and games. Through these activities, children learn to categorize, understand, and generalize

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the world around them. Play allows the development of discovery, exploration, and abstract thinking. All of which help in concept formation. In this context, play and development-related activities support learning continuity.

This chapter explains how children create their conceptions and focuses on the importance of concept development in the learning process. In simple words, a concept is the main idea of understanding something, which tells us what a thing is and how it works. For example, when we say, “A tree is a plant that is big and has a trunk, branches, and leaves,” this is the concept of a tree. Another example: When we talk about a family, the concept is that a family is a group where parents, siblings, and other members live together and support each other. Concepts often develop gradually in infants and young children as a result of play and environmental involvement. The formation of concepts must be encouraged in order to improve their cognitive abilities. Children begin to classify things in their early years, and by middle childhood, they develop deeper conceptions. It has also been noted that they actively construct ideas from what they already understand. They continuously modify their existing thoughts and make connections between new and current as well as previous knowledge as they learn.

**6.2 LEARNING OUTCOMES**

After completion of this chapter, the learner:

- identifies ways to use children’s existing knowledge for introducing and constructing new concepts;
- describes the process of concept learning in children through their engagement in different roles, daily life routines, and interactions within home, school, and environmental settings;
- implements the Panchpadi (five-step) pedagogical process to facilitate meaningful concept development in children;
- develops a graphic organiser or concept map that visually represents children’s connection of ideas from home, school, and their surroundings.

6.3 NATURE OF CONCEPT

Concept means grouping objects, events, or ideas based on common features and organizing information into categories. The development of concepts starts in childhood. Children learn through their senses of taste, smell, sight, hearing, and touch. Children learn about size, weight, and shape by observing their surroundings. Shapes of objects are learned from the experiences of touching. A child's understanding of the world is influenced by this process which starts early in infancy and continues throughout childhood. A more detailed look at how concepts are formed in childhood is provided here:

Sensory Experiences: Using their senses of touch, smell, sight, hearing, or taste, infants begin building concepts.

Experiences and Interactions: Through their interactions with others and their surroundings, children gain knowledge about the world.

Isolating Features: By using similar characteristics, they can identify unique characteristics of things and events.

A young child looks at the family pet cat and sees that it has four legs and fur and makes a meow sound. Later, while visiting a park, the child also sees other cats that have similar characteristics: four legs, small tails, and meows. Over time, the child begins to form a mental concept that all cats have four legs and meow sounds. One day, the child sees a dog and excitedly calls it a "cat" because it also has four legs and fur. A parent gently corrects the child, explaining that although cats and dogs share some characteristics, dogs bark and have different shapes and behaviours, while the same cat makes meow sounds and has different behaviour. Through this experience, the child modifies his/her understanding, learns to differentiate between similar animals, and develops a clear concept about what defines a cat.

Definitions of Concepts

1. **Ross:** "A concept is the mental construction of an idea."
2. **Boring, Langfeld, and Weld:** "Concepts are ideas that refer to objects, events, qualities, etc."

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6.4 PROCESS OF CONCEPT FORMATION IN CHILDREN

According to **Jerome Bruner**, a child has to pass through five stages in concept formation which are as follows:

1. **Comparison:** Through observation, the child creates two concepts of horse in his mind - one white and one black. After that, he compares those two concepts. He has seen two horses. Both are of different colors. Despite this difference, he finds similarity in them.
2. **Generalisation :** Due to the accumulation of similar qualities, there is no difference for the child between horses of different colors. Hence, his concept (image) of a horse takes a clear form. Now he calls any horse a horse. In this way, he gets knowledge of the horse species.
3. **Observation:** The child starts forming concepts even before learning to speak. First, he sees many objects and forms concepts or mental images of them. For example, he sees a white horse. As a result, he gets to know its concept. In other words, he forms its image by observing the horse. After some time, he sees a black horse. By observing it, he forms its concept as well.

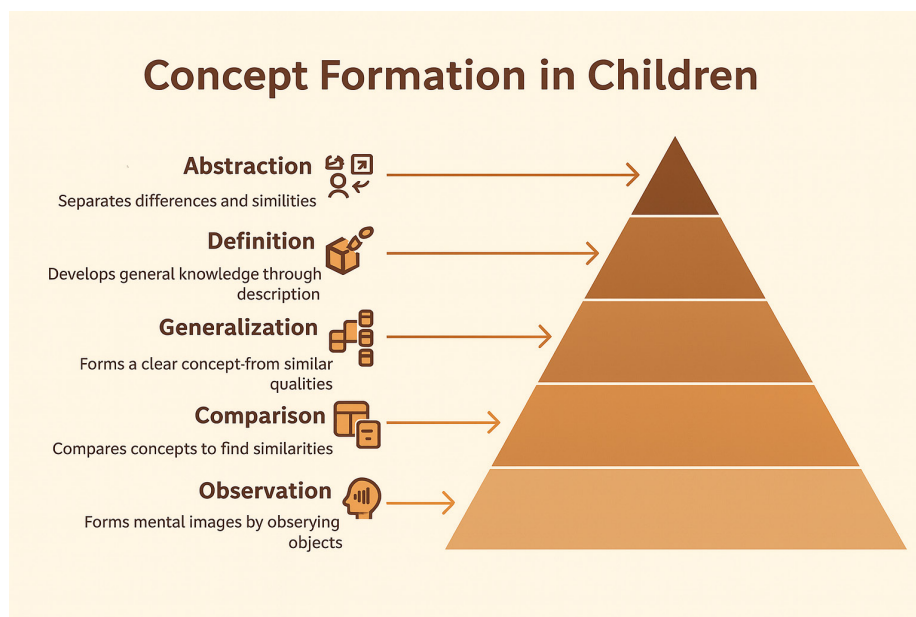


Fig. 6.1: Concept formation process (AI - Generated Image)

4. **Definition Formation-**After passing through the above three stages, the child forms the concept of horse. This concept gives him general knowledge of horses. We can give him this knowledge only through description or

definition. When we describe a horse to him, he thinks about a horse in general by thinking about a particular breed. This is the real concept that has been formed in his mind.

5. **Abstraction:** The child separates the differences and similarities between the two horses. The difference is only due to the color. The similarity is due to many things. He adds these similarities or similar qualities to the differences.

Once a child forms a deep concept (for example, “bird”), they can apply it to new examples and begin to build more complex ideas, such as the life cycle of birds, their habitats, and more. In this way, concept formation becomes a gateway to broader learning across all subjects.



CHECK YOUR PROGRESS 6.1

1. Define the word “concept” in your own words, with an example.
2. Mental categories that are used to group objects, events, and information are called:
 - a) Concept
 - b) Connector
 - c) Experience
 - d) Generalization
3. Who said “A concept is the mental construction of an idea”?
 - a) Ross
 - b) Boring
 - c) Bruner
 - d) Piaget

Ans: 2-a; 3-a

6.5 TYPES OF CONCEPTS

After understanding the concept’s nature, it is important to know its various

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forms. During their entire lives, children develop a wide variety of ideas. Some of these are described here:

1. **Concrete Concepts:** These are concepts that can be directly experienced through our senses, such as “apple” and “dog”. These are relatively easy to learn and remember because these objects are real and tangible.
2. **Abstract Concepts:** Abstract concepts are ideas or qualities that cannot be directly seen or touched, such as “freedom” and “love”. To understand these, more complex and higher level mental processes are required.
3. **Natural Concepts:** These concepts are formed from our real-life experiences. Their boundaries are not clear and can change according to the context, for example - “furniture” (which may include chairs, tables, etc.). These are learned from experience and environment rather than formal education.
4. **Artificial Concepts:** These are concepts that are defined based on particular rules or characteristics, such as – geometric shapes: “triangle” and “square”. These are clear and precise and there is less variation in their meaning.
5. **Prototypes:** Every concept has a best example or “ideal form” to which other things are compared. For example, the concept of “bird” might have “sparrow” or “robin” as an ideal example. These help us in classification.
6. **Schematic Concepts:** These are mental frameworks or models that help us organize our experiences and information, such as “scripts” for social interactions. These help us to think quickly in different situations and predict possible outcomes.
7. **Categories:** Categories are a way of dividing concepts into groups or hierarchies. This makes information easier to organize and remember. These are formed on the basis of common properties or functions.

According to Jerome Bruner, there are three types of concepts-

1. **Conjunctive Concepts**— These are concepts where one or more features are the same in a group of objects. For example, a bird, cow, and dog belong to a group because they are all animals. Similarly, a chair, table, and bed can form a group because they are all pieces of furniture.

2. **Disjunctive Concepts**—These are when some things have similarities, but not everything is the same. Example: Tea and coffee are both hot drinks, but they differ in taste and how they are made.
3. **Relational Concepts**—These are when we understand things by comparing them with something else. Example: A tall building looks taller when we compare it to a shorter building.

Example: Chairs have many designs, styles, and shapes, yet when you see different styles of chair, you know that they are chairs because you have the concept of a chair in your mind. Another example is the concept of clothes. Clothes may include dresses, shirts, jeans, shorts, etc. Despite these differences, anyone can easily recognize what items are clothes and what are chairs. This is because of clear concepts of what chairs and clothes should be.



CHECK YOUR PROGRESS 6.2

1. Explain types of concepts using real-life examples in an interactive format.
2. Which of the following statements is correct about concept formation by a child?
 - a) Concepts are one dimensional
 - b) Concepts are hierarchical
 - c) Concepts can be both simple and complex
 - d) The process of conception occurs only in children
3. Types of Concepts by Jerome Bruner
 - a) Three
 - b) Five
 - c) Three
 - d) Four

Ans. 2-c; 3-c

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6.6 THEORIES RELATED TO CONCEPT FORMATION IN CHILDREN

Children develop concepts in many ways, and various psychologists have proposed theories to explain this process. These theories help us understand how new concepts are formed in children, and the stages involved in concept development. These theories are as follows:

6.6.1 Jerome Bruner's Theory of Cognitive Development

Jerome Bruner, a renowned psychologist, made significant contributions to understanding children's cognitive development. He explained how children learn and acquire knowledge. The main points of Bruner's theory are:

1. Three Modes of Representation

According to Bruner, children learn through three stages:

- **Enactive Representation (Birth to 1 year):**

Learning occurs through actions. Knowledge is stored in the form of physical activities (e.g., grasping an object, walking). For example, a child learns to shake a rattle to produce sound. Even when the toy is taken away, the child may repeat the shaking action, expecting the same sound.

- **Iconic Representation (1 to 6 years):**

Learning occurs through images and mental pictures. Thinking is based on visual cues. Some individuals can consciously visualize images, while others do so subconsciously. For instance, a child learns to recognize a mango by looking at its picture. Visual aids like images, charts, and diagrams help in understanding new topics.

- **Symbolic Representation (7 years and above):**

Learning occurs through language and symbols (e.g., words, numbers). This is the most complex form and marks the beginning of logical thinking. For example, a child can understand mathematical expressions like $5 + 3 = 8$. The word "dog" symbolizes a particular category. Symbols can be sequenced, classified, and reorganised.

2. Spiral Curriculum

Bruner believed that any subject can be taught to children at any age if presented appropriately. Therefore, the curriculum should be designed so that students revisit topics repeatedly with increasing depth—this is known as the “spiral” approach. For example, a child learns basic concepts of energy in primary school and explores the same topic in more depth in higher grades.

3. Discovery Learning

Bruner emphasized that children should actively participate in the learning process. When children discover knowledge themselves, it becomes more lasting and effective. Teachers should act as guides rather than mere information providers, helping students find connections between pieces of information.

Bruner’s theory explains how children develop knowledge—starting with actions, then through images, and finally through symbols. He promoted “discovery-based learning” and the “spiral curriculum,” which remain significant in educational methods till today.

6.6.2 Piaget’s Theory of Cognitive Development

Jean Piaget, a Swiss psychologist, described how children think and learn. He divided cognitive development into four stages:

1. Sensorimotor Stage (Birth to 2 years)

Children learn through their senses (seeing, hearing, and touching) and actions. They develop “object permanence”, an understanding that objects continue to exist even when it is not visible.

2. Preoperational Stage (2 to 7 years)

Children represent things using symbols (words, images). They exhibit “concentrated thinking”, focusing on one aspect of a situation. They are “egocentric”, unable to understand others’ perspectives.

3. Concrete Operational Stage (7 to 11 years)

Children understand logic and rules through concrete examples. They grasp the concept of conservation, recognizing that quantity remains the same

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despite changes in shape. They can classify and sequence objects.

4. Formal Operational Stage (12 years and above)

Children can think abstractly and logically. They develop problem-solving and imaginative thinking abilities.

6.6.3 Vygotsky's Sociocultural Theory of Cognitive Development

Lev Vygotsky, a Russian psychologist, believed that a child's cognitive development is influenced by social and cultural contexts. His main concepts include:

1. Sociocultural Influence

Children learn not in isolation but through interactions with people around them (parents, teachers, peers). Language and culture play crucial roles in learning.

2. Zone of Proximal Development (ZPD)

This is the gap between what a child can do independently and what they can achieve with assistance. With proper guidance, children can perform more complex tasks.

3. Scaffolding

This is a supportive process where a teacher or experienced individual guides a child through difficult tasks. As the child learns, this support is gradually reduced.

4. Role of Language

Vygotsky believed that language is not just a means of communication but also a tool for thinking. Internal speech helps children plan and solve problems.



ACTIVITY

Recognise and describe Piaget's four stages of cognitive development using role-play and real-world examples.

**CHECK YOUR PROGRESS 6.3**

1. What is the main objective of the spiral curriculum?
 - a) To repeat subjects
 - b) To teach subjects in a sequential manner
 - c) To teach subjects repeatedly with increasing depth
 - d) To teach difficult subjects only in higher classes
2. Which developmental stage is characterized by “egocentric thinking”?
 - a) Sensorimotor
 - b) Preoperational
 - c) Concrete Operational
 - d) Formal Operational
3. Which stage of development does Piaget believe is characterized by abstract thought and advanced imagination?
 - a) Formal Operational
 - b) Concrete Operational
 - c) Preoperational
 - d) Sensorimotor
4. What does ZPD (Zone of Proximal Development) represent?
 - a) A child’s IQ level
 - b) A gap between what a child can do independently and with assistance
 - c) Language skills
 - d) Emotional understanding

Ans. 1-c; 2-b; 3-a; 4-b

6.7 CONCEPT FORMATION IN CONTEXTS OF LEARNING: HOME, SCHOOL, ENVIRONMENT

After studying various theories of concept development, we can now clearly understand how children form concepts. Children build and learn concepts in

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every context of their lives. The three main learning contexts are home, school, and the surrounding environment, each play important role in the concept development process.

6.7.1 Learning Concepts through Experiences at Home and with Family

Home is the primary environment where children begin to develop concepts about their immediate world, objects, actions, and relationships. From birth, home becomes the first learning place where parents and family members provide countless informal learning opportunities.

1. **Daily Routines:** Children can learn about materials, measurements, and creativity by applying principles to daily tasks like playing or cooking. Observing or participating in cooking helps teach science and math topics like amount, hot and cold, and taste. Concepts can also be developed by counting utensils or discussing fruits and vegetables during meals. Saying “Good morning” or “Good night” helps develop concepts, so even small rituals count.
2. **Role-play:** Pretending to ride a bus or act as a parent enhances concept development through imagination.
3. **Language and Communication:** By listening to adults talk and participating in conversations, children learn social rules, relationship values, and behaviour. Rich language and clear explanations from parents and caregivers help children understand and use language better, aiding their cognitive and emotional development.
5. **Sharing Responsibilities with Family:** Tasks like cleaning the room or watering plants teach children responsibility and discipline.

Thus, home supports concept development through interaction, play, and routine, offering diverse examples and real-life exposure. Language develops through local stories and conversations, and early numeracy skills are built through home activities like cooking or shopping.

6.7.2 Learning Concepts through Experiences at School

Schools provide structured opportunities for concept development through curriculum, lessons, and interaction with teachers and peers. They also offer a social space where children learn to interact, empathise, and manage emotions—crucial for overall concept development.

1. **Real-World Connections:** Schools link learning to real-life situations, helping children apply knowledge and skills practically.
2. **Critical Thinking:** Schools promote critical thinking by encouraging children to analyse information, express opinions, and participate in discussions.
3. **Discussion and Interaction:** Talking and discussing with teachers and classmates enhances language, manners, and social skills. Classroom conversations also matter, e.g., if two children say “pen” and “pan” for the same object, the teacher’s guidance helps form the correct concept.
4. **Group Activities and Games:** Activities like teamwork and following game rules help children understand cooperation and handle both winning and losing positively.
5. **School Routines:** Daily school routines like prayer, walking in a line, or lunch break teach discipline, time management, and social habits.

Concepts learned at school include discipline, cooperation, task division, and social skills.

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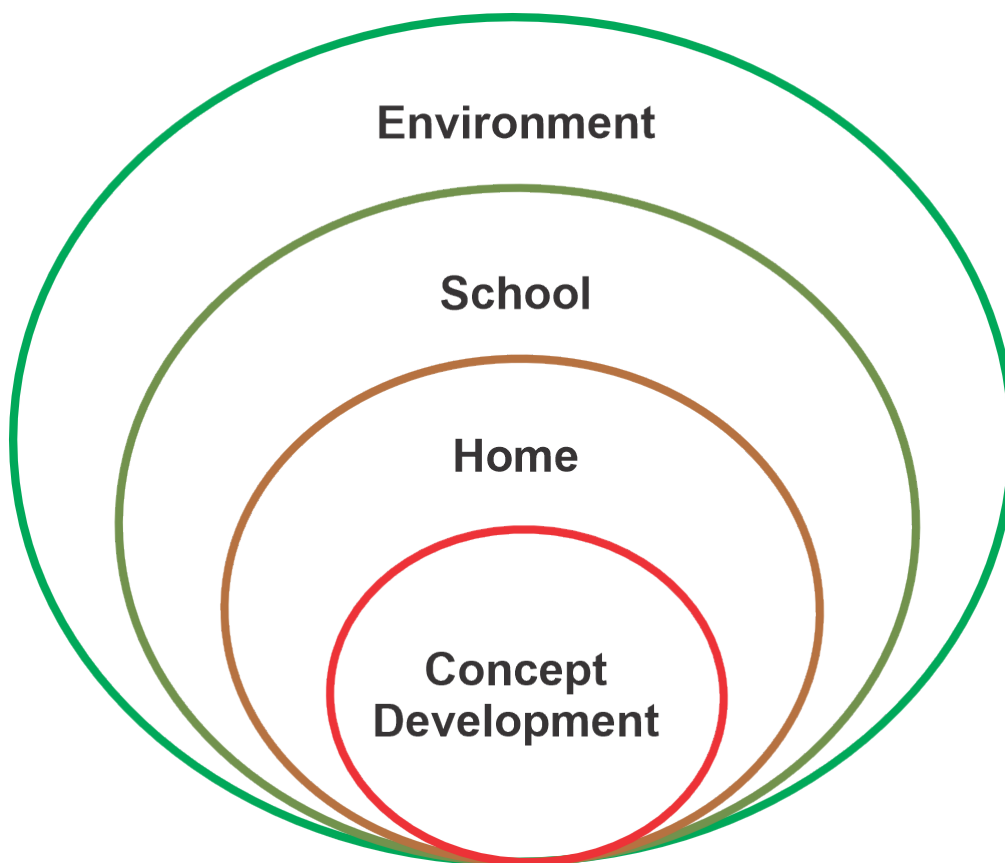


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**Fig. 6.2: Concept Formation in Contexts of Learning:
Home, School, Environment**

6.7.3 Learning Concepts through Environmental Experiences

The environment provides real-world contexts that help build concepts that may not be easy to teach in the classroom. Experiences like gardening, shopping, or zoo visits help children form clearer concepts than pictures alone. For instance, seeing real farm animals (cow, goat, hen) helps children classify them better.

1. **Broader Exposure:** Community programs, museums, and trips expose children to different cultures, ideas, and perspectives, broadening their understanding of the world.
2. **Environmental Awareness:** Exposure to parks and nature centers helps children develop ecological, biodiversity, and conservation concepts.
3. **Life Skills:** Environmental experiences promote the development of important life abilities or skills that are required for success in future, such as organization of time, making decisions, and problem-solving abilities.

- 4. Connecting Knowledge:** Children learn to link what they've learned at home and school with real-life experiences, strengthening their understanding and application of concepts.

Experiences in markets, parks, temples, or buses teach real-world complexity, diversity, and rules.

**CHECK YOUR PROGRESS 6.4**

1. Name any two ways how home helps children in concept formation.
2. Mention a classroom activity or routine that helps children build concepts.
3. Where do children start learning concepts first?
 - a) School
 - b) Tuition
 - c) Home
 - d) Library
4. How do peer discussions and teacher guidance help children?
 - a) In storytelling
 - b) In forming correct concepts
 - c) In winning games
 - d) In solving math problems
5. Choose correct True/False Statements
 - a) Children learn language and social rules from family.
 - b) Schools only provide bookish knowledge, not practical experience.
 - c) Environmental visits help children understand complex concepts practically.
 - d) Playing games distracts children from concept building.

Ans. 3-c; 4-b; 5-a True; b) False c) True d) False

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6.8 FACILITATING ACTIVITIES RELATED TO CONCEPT DEVELOPMENT IN CHILDREN

After understanding the different theories of concept formation, now we will discuss how children develop concepts. A child begins to form concepts when they can see a connection between new and past experiences and understand the meaning behind them. Prior knowledge acts as a bridge between what is already known and what is yet to be learned. This connection makes it easier for children to grasp new information and encourages deeper learning.

Teachers play an important role in this process. In classroom, they should help children compare things using examples. For example, they can ask, *“How are these two plants similar and different?”* It helps children understand how both are related to the same concept. Example: If a child says, *“The moon is following me,”* the teacher can gently understand the child’s thinking and gradually correct it and explain the scientific reason behind it. After understanding how concepts develop in children, it’s important for teachers to know how to introduce and facilitate new concepts in children. Here are some effective strategies:

1. **Use Concrete Examples and Comparisons:** It is effective to begin teaching new things by starting with what they already know. When teachers connect new concepts to familiar words or ideas, learning becomes easier and more meaningful.

For example, if a child already understands what a “circle” is, it becomes easier to explain the concept of a “wheel.” Similarly, introducing a story, song, or familiar example (like a counting rhyme before teaching addition) helps prepare the child’s mind for learning the main topic.

2. **Encourage Connections and Meaning-Making:** To help children understand, teachers should ask questions and encourage discussions that boost thinking. For example, when a teacher shows pictures of fruits and asks, *“What is similar about these fruits?”* Children may answer, *“They are all sweet and grow on plants.”* This helps them understand characteristics of fruits. Stories and comparisons also help children learn new things. For example, to explain “herbivore,” the teacher can compare a cow (eats grass) and a lion (eats meat). A simile like *“A car’s engine is like its heart”* makes complex ideas easier to understand. Teachers should also encourage children to make their own comparisons to improve thinking. Later, when the teacher connects big ideas like “photosynthesis” or “life cycle,” they

can draw a map on the board to show how everything is connected. This helps children learn the concepts in a clear manner.

3. **Help Recognize Relationships:** To help children understand relationships and key concepts, especially in subjects like social studies, teachers should use clear examples and comparisons. For instance, when teaching about classification, a teacher can create a chart comparing village and city life—for example, villages have farms, while cities have factories. The teacher can then ask, *“What is similar and what is different between villages and cities?”* This approach helps children grasp differences, similarities, and cause-effect relationships.

If the topic is “road safety,” show a picture of a person wearing a helmet and one without it, and ask, “Who is safer and why?” This helps children learn the right behavior. Linking concepts across subjects is useful too. While learning about “rivers” in social studies, ask them to draw a river in art class or use the word “river” in a poem. This helps children understand one concept in many ways and connect it across subjects.

4. **Questioning Technique:** Ask open-ended questions based on children’s experiences. Example: Topic: Weather Question: *“Have you ever enjoyed getting wet in the rain?” “What do you wear in summer?”* Explain the characteristics of the seasons from children’s experiences.
5. **Storytelling and Sharing Personal Experiences:** Encourage children to share their stories or experiences related to a topic. Example: Cleaning and hygiene Question: *“Do you brush your teeth every day?”* Following this, tell a story like, *“Chiku, a rabbit did not brush his teeth and got cavities...”*
6. **Use Real-Life Situations:** Explain new and abstract topics with familiar daily life examples. Example: Teach addition and subtraction while shopping or playing. Example: Topic: Addition/Subtraction. Ask in class, *“If you have 5 toffees and you give 2 to your friend, how many are left with you?”*
7. **Brainstorming Session:** Allow children to express their views freely about what they know about a topic. This will reveal their level of understanding and clear any misconceptions. Example: Topic: Animals. Ask children, *“Which animals do you know?”* Write down students’ answers on the board. Then make them classify animals into domestic and wild animals.

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8. **Group Discussions and Peer Teaching:** Children learn from their peers how they connect a topic with their previous knowledge. This promotes collaborative learning. Example: Topic: My family. Divide children into groups of 2–3. Each child should talk about his/her family members. Then children should compare among themselves— *“How many members are there in your family?”*
9. **Use Visuals and Multimedia:** Use pictures, videos, or diagrams that connect old knowledge to new topics. Example: Use a picture of a tree to explain both the parts of a plant and the water cycle. Topic: Water cycle. Activity: Show children pictures of clouds, rain, and sun, and play a simple video. Then ask, *“How did you feel the rain?”* and add how the sun makes water evaporate and clouds form.
10. **Create New Beliefs based on Wrong Beliefs:** Identify children’s incorrect prior knowledge and correct it gently. Example: If a child believes that “the sun moves,” make this the basis for explaining the rotation of the earth. Topic: Day and night. Some children may say, “The sun rotates.” Correct: *“The earth rotates, which causes day and night.”* Demonstrate an activity with a globe and torch.
11. **Create a Supportive Learning Environment:** Create a supportive and positive learning environment for children by encouraging them, answering their questions, and providing support when needed.
12. **Encourage their Natural Curiosity:** Foster children’s natural curiosity by encouraging them to ask questions, explore their surroundings, and seek out new information.

Using these methods can help a child build strong concepts and develop a love of learning that will last for lifetime.

**ACTIVITY**

Explain two strategies and plan it in the way which you can use to connect new concepts to the concepts already known to children.



CHECK YOUR PROGRESS 6.5

1. What should be done to correct the misconceptions of children?
 - a) They should be scolded
 - b) They should be ignored
 - c) Wrong knowledge should be identified, and scientific explanation should be given
 - d) Give them homework
2. How does using prior knowledge help in understanding new information?
 - a) Forgetting the information
 - b) Making it difficult to remember
 - c) Linking new information to past experiences
 - d) Decreasing interest in studies

Ans. 1-c; 2-c

6.9 ACTIVITIES TO FACILITATE CONCEPT DEVELOPMENT IN CHILDREN

Some of the activities to facilitate concept development are as follows:

1. **Hands-on Projects:** Activities like building with blocks, building towers, or creating art projects help children learn and understand concepts in a fun and interactive way.
2. **Outdoor Games:** Encouraging children to play outside and engage with their environment can help them understand scientific concepts such as weather patterns and seasons.
3. **Cooking:** Cooking and baking can help children learn about math, science, and nutrition in a fun and engaging way.
4. **Science Experiment:** Simple science experiments can help children understand basic concepts related to physics, chemistry, and biology.

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5. **Nature Walk:** Going on walks in nature and observing the environment can help children understand concepts related to biology and ecology.
6. **Art Projects:** Art projects can help children explore and understand concepts related to creativity, self-expression, and history.
7. **Chorus Movements and Dances:** Participating in these activities can help children understand concepts related to music, rhythm, and coordination.
8. **Debate:** To encourage students to engage in lively discussions and debates to develop their critical thinking and communication skills.
9. **Inquiry-Based Learning:** Encouraging students to ask questions, gather information, and explore concepts through practical learning experiences.
10. **Problem-based Learning:** Using real-world problems as a starting point for learning, exploring concepts, and finding solutions.
11. **Games and Puzzles:** Design games or puzzles that involve the concept as part of the challenge. This makes learning fun and interactive.
12. **Role-Playing:** Let children act out scenarios related to the concept. This helps them understand the different roles and perspectives involved.

**ACTIVITY**

Design a practical activity that can help build a concept (from any subject).

**CHECK YOUR PROGRESS 6.6**

1. Which of the following activities is suitable for promoting concept formation in children?
 - a) Teaching only the textbook
 - b) Practical projects
 - c) Oral exam
 - d) Dictation

2. Choose the correct/wrong statement from the following

- a) Engaging children in debates and discussions develops their critical thinking.
- b) Only classroom teaching is sufficient for concept formation.
- c) Inquiry-based learning motivates children to explore concepts.
- d) Games and puzzles are boring ways for children to understand concepts.
- e) Science experiments help kids understand concepts that are simply related to art.

Ans. 1-b; 2- a) True b) False c) True d) False e) False

6.10 USES OF GRAPHIC ORGANISERS AND CONCEPT MAPS

Visual connectives are powerful tools for learning concepts. Graphic organizers such as charts, Venn diagrams, timelines, and concept maps are used to help children visualize concepts and the relationships between them to externalize their ideas. It also helps assess children's conceptual development and understanding, identify misconceptions, and facilitate learning by building new knowledge on old knowledge. Graphic organizers and concept maps are visual representations of knowledge, where concepts are represented as nodes, and the relationships between them are connected by lines or arrows.

Concept mapping is made up of two words: concept and mapping. In simple terms, a concept is an idea, and a map is an image that represents part or all of something. This means that a concept map represents different parts of an idea. Concept maps were first created by Joseph Novak in 1972, and he found that concept maps can be effectively used in learning processes, especially for children. He found that it helps in organizing information, identifying relationships between concepts, and understanding how knowledge is interconnected.

Graphic organizers and concept maps help children learn by:

1. **Visual Organization:** These visual coordinators provide a visual structure for organizing information, making it easier for children to understand complex ideas.

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2. **Identification** of these help children understand how different concepts are connected and related to one another.
3. **Meaningful Education:** Promotes meaningful learning by focusing on relationships between concepts rather than rote learning.
4. **Active Study:** Creating these visual connectors involves children actively organizing and connecting information, which makes the learning process more engaging and memorable.
5. **Assessment:** These visual coordinators can be used as an assessment tool to evaluate student's understanding of a topic.
6. **Differentiation:** These visual coordinators can be adapted to suit different learning styles and needs, making them a versatile tool for all learners.

These visual connectors change the way of learning, thinking, and communicating ideas. By connecting concepts visually, students and teachers move beyond memorization to deeper understanding. Instead of reading long paragraphs or lists, these visual connectors present ideas in a way that is easy to see and remember things. When connecting concepts visually, the brain processes and retains information better and making studies more effective. Some subjects may seem overwhelming to children, but visual connectors help simplify learning by breaking down large ideas into smaller connected parts.



Fig. 6.3: Steps to create a concept map

Steps to create a concept map:

1. **Identify the main topic:** - Select the central concept or question you want to study.
2. **Consider the main concepts.** Brainstorm and list out the topic. Write down all relevant words and ideas related to the main topic.
3. **Organize the concepts:** Make a group of related concepts and organize them visually.
4. **Draw lines and connect the ideas:** Connect concepts with lines or arrows to show relationships between them.
5. **Add labels:** Use labels to explain relationships between concepts.
6. **Review and refine:** Revise and refine your map, adding more connections and details as needed.



CHECK YOUR PROGRESS 6.7

1. Discuss how concept maps help in organizing, connecting, and remembering new ideas
2. Who created the concept map and when?
 - a) John Dewey, 1950
 - b) Joseph Novak, 1972
 - c) Sigmund Freud, 1960
 - d) Piaget, 1980
3. How are concepts represented in a concept map?
 - a) In the form of diagrams
 - b) In the form of nodes
 - c) In the form of paragraphs
 - d) In the form of tables

Process of Learning



Notes

Process of Learning



Notes

4. What is used to show relationships between concepts in concept maps?
- Picture
 - Colour
 - Lines or arrows
 - Graph

Ans. 2-b; 3-b; 4-c

6.11 PANCHAPADI METHOD IN CHILDREN'S EDUCATION

The Panchpadi method is an effective teaching process which is a five-step pedagogical approach recommended in the National Curriculum Framework (NCF) 2023 for foundational stage education. It aims to tailor teaching to the child's interests, understanding, and experiences so that they can understand concepts deeply and intuitively. Panchpadi has five interrelated steps:

- 1. Introduction (Aditi):** The aim is to generate curiosity and connect the new knowledge to the child's existing knowledge. When new topics are introduced in a meaningful way and in a relevant context and lesson is started with an interesting activity, story, question, or example, it generates interest in the child. Example: If the topic is "Plants," the teacher asks, *"Have you seen any plants in your home or park?"* Children share their experiences: *"I have seen Mummy watering the plants."*
- 2. Conceptual understanding (Bodha):** Helps children understand basic ideas through interactive and sensory-rich activities. Concept formation is built through hands-on experiences. Children actively construct knowledge rather than passively receive it. Children's understanding deepens when they touch, see, and do things themselves. Example: Children plant seeds, watch them grow every day, and identify the parts of a plant (root, stem, leaf, flower). This helps them clearly understand "how a plant grows."
- 3. Practice (Abhyasa):** It aims to reinforce learning through repetition. Practice helps children consolidate their understanding by repeating the concept through different activities. Repetition and varied practice of any concept and thing strengthen the neural connections, making the concept more stable and accessible. For example, children can engage in games like matching leaves to plant parts, drawing plant parts, singing songs about how

a seed grows, or dramatising the life cycle of a plant. This repetition across formats ensures that the concept gets embedded in long-term memory.

Process of Learning



Notes

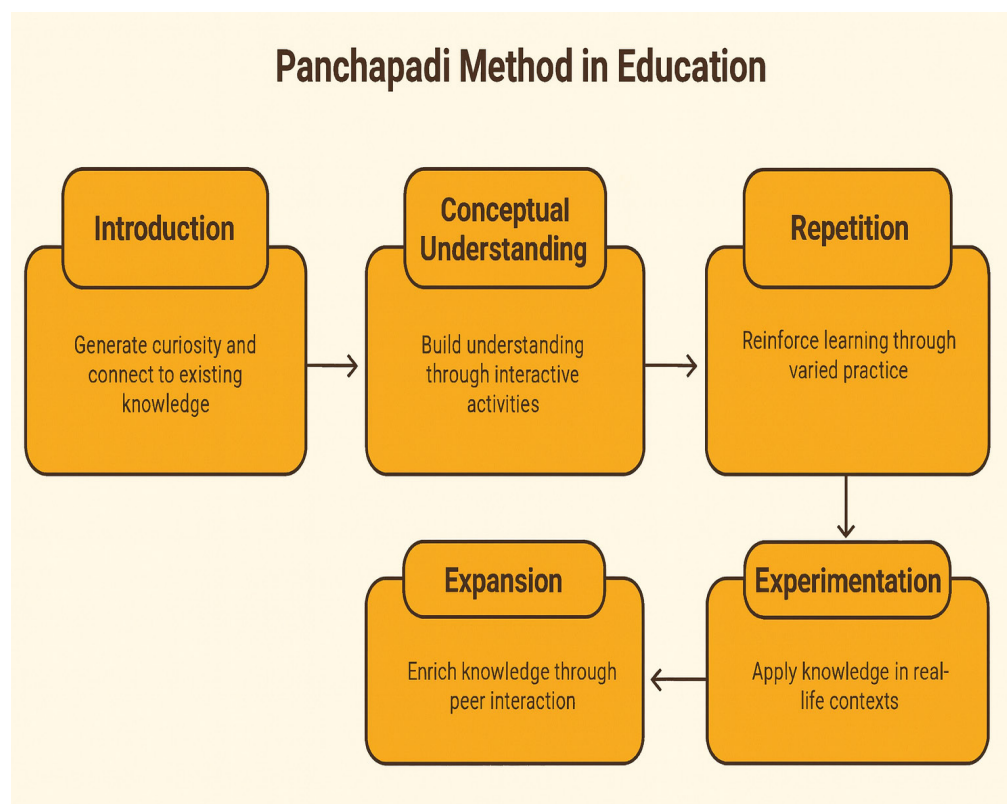


Fig. 6.4 : Panchapadi Method in Education

4. **Application (Prayog):** Its purpose is to help children transfer their understanding to real-life contexts. Applying knowledge in new contexts leads to concept generalisation. Children understand that concepts are not limited to the classroom. For example, children may be asked to take care of the school garden or identify which plants bear fruits or flowers. They can sort out vegetables at home or water the plants at home. Through this, they realize that their learned knowledge is applicable in real world.
5. **Expansion (Prasar):** It aims to enrich the range of knowledge by expressing and learning from peers. When children discuss, explain, or communicate what they have learned with their peers, family, and other social groups, this leads to deeper understanding and thinking. For example, children can tell a story about a plant's journey from seed to flower, sing a song, and work together to make a group poster on "How Plants Grow." They also listen to the experiences of others and add to their mental model of the concept.

Process of Learning



Notes

Table 6.1: Use of Panchapadhi method to teach the topic “Water Cycle”

Panchapadhi Steps	Activities
1. Introduction	a) Ask the children, “ <i>What will happen if the rain stops?</i> ” Show a short video clip.
2. Conceptual understanding	b) Explain the water cycle to children by drawing a picture or telling it in the form of a story.
3. Practice	c) Revise through quizzes, practice assignments or group discussions
4. Application	d) Consider using rainwater harvesting
5. Expansion	e) Ask children, “ <i>How does water vapor form?</i> ” and “ <i>How do clouds form?</i> ”

The Panchapadhi Teaching Method makes the learning process natural and joyful. It enhances children’s deep understanding, meaningful application, and social sharing. The process supports whole-brain development, especially in children’s early years, by integrating thinking, action, feeling, and communication which are essential for strong concept development.

**ACTIVITY**

Comprehend and demonstrate the five steps of Panchapadi with examples through any subject topic.

**CHECK YOUR PROGRESS 6.8**

Match each step of the Panchapadi learning process with its corresponding activities.

Pancpadi Steps	Related Activities
1. Aditi	a) Practice and reinforcement of the concept.
2. Bodh	b) Applications in real life or new contexts.
3. Abhyasa	c) Understanding the concept through guided activity.
4. Prayoga	d) Extension and sharing of learning.
5. Prasar	e) Introduction using prior knowledge.

Ans. 1-e

2-c

3-a

4-b

5-d

**6.12 LET US SUM UP**

Children learn concepts through experience and observation. At home, children acquire knowledge from their parents and relatives through daily activities, games, stories, and conversations. Teachers in school use games, examples, collaborative work, and other methods to demonstrate new concepts and contribute to students' existing knowledge. In contexts such as parks, marketplaces, temples, or zoos, youngsters gain a deeper understanding of things via observation. They are used to facilitate the learning of concept maps (mind maps) and graphic organisers (charts, pictures). The Panchpadi style of instruction begins with a scenario, followed by discussion, practice, reflection, and questioning. This approach encourages understanding and thoughtfulness in children. Applying theories of concept development makes it simpler for students to understand how concepts are formed.

According to development theorist Bruner, children learn by doing (like playing), then by seeing pictures, and finally by hearing and seeing words and symbols. Additionally, he stated that if a material is presented properly, children can learn it in accordance with their developmental stage. Piaget separated the cognitive development of children into four phases. Young children learn by observation, followed by thought, and reason. According to Vygotsky, children learn best when they engage with and get support from others. His “ZPD” concept explains what children might achieve on their own and with guidance. Children progressively acquire new ideas through their experiences, queries, and ideas. This way makes learning exciting experience.

**6.13 END EXERCISES**

1. Define concept formation and discuss its importance in early childhood learning.
2. Describe three strategies that a teacher can use to facilitate concept formation in a classroom of young children. Provide an example activity for each strategy.
3. Explain how home environment and school environment contribute to concept formation. Give a specific examples.

Process of Learning*Notes*

Process of Learning



Notes

4. How can children learn new concepts with the aid of graphic organizers like concept maps or charts?
5. Explain the Panchapadi approach. Explain how you would use each of the five Panchapadi steps (Aditi, Bodh, Abhyasa, Prayog, Prasar) to teach a basic concept (like shapes, animals, or parts of speech).

6.14 REFERENCES

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